SectraTEK

AI FOR CYBERSECURITY AND DIGITAL FORENSICS · LAB 08

Ask the Bug Report



A forensic chatbot over an Android bug report: ask in plain English, and get an answer that cites the section it came from. Use a report from your own phone.

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THE IDEA

From a whole extraction to a single text file

A full phone extraction is large and needs special tooling. An Android bug report is just a text file, so any student can make one and practise asking a model grounded, cited forensic questions.

A FULL EXTRACTION

Gigabytes, special tools

Powerful, but heavy to acquire and store. Hard to hand to every student.



A BUG REPORT

One sectioned text file

Lightweight, and you can pull one from your own phone with a single command.

THE REPORT

What is inside an adb bug report

The output of `adb bugreport` is one text file split into labelled sections. The ones that matter here:

Device

Model, Android version, build: the device fingerprint.

Packages

The apps installed on the device.

Accounts

The accounts configured on the device.

Wi-Fi

Saved networks and connection state.

Battery

Charge level and power state.

System log

Recent events, errors, and crashes.

THE PROBLEM

Ask a model directly, and it invents

"This phone has Gmail, WhatsApp, and a banking app installed."

WHY THAT IS A PROBLEM

It never saw the report

The model answers from general training, not this device.

It invents specifics

Plausible app names and settings that are not in the file.

Nothing to check

No section is named, so the answer cannot be verified.

The fix is to give the model the report and read-only tools, so it answers from the file and cites it.

THE AGENT

Three read-only tools over the report

The model never reads the whole file blindly. It is given three tools and decides which to call.



```
list_sections()
```

List the report's sections, so the agent knows what is available.



```
read_section(name)
```

Read one section, by number or by text in its title.



```
search_report(keyword)
```

Search the whole report; return matching lines with their section.

Read-only: the agent can list, read, and search, but never change the report.

HOW IT WORKS

Think, act, observe, then answer with the section

Question

A plain-English forensic question.

Think + act

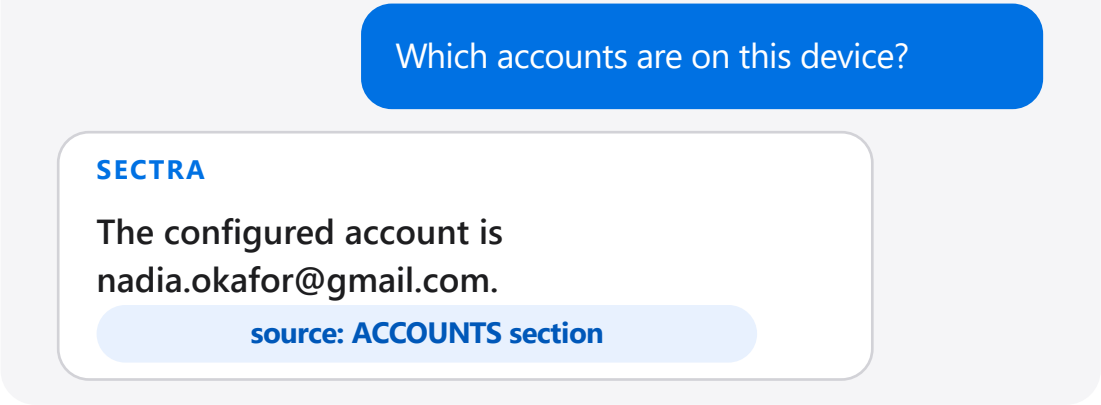
Pick one tool and call it.

Observe

Read what the tool returned.

Answer

Reply, and cite the section.



It loops until it can answer

Search, read, repeat: one tool per step.

It cites the section

Every answer names where it came from.

You can verify it

Open that section and check for yourself.

THE EVIDENCE CHECK

The code checks every answer first

Read-only tools limit what the AI can look at, not what it can say. So before you see an answer, the code checks it against what the tools really returned.

Did it look?

At least one tool must have run for the question. A reply that only asks you a question back is exempt.

Is every detail real?

Every file, IP address, process ID, and memory address must appear in what a tool returned.

Is the citation honest?

The answer cannot cite a tool that never ran.

IF IT FAILS

It goes back to the AI to check with a tool or drop the claim. After two tries it arrives marked UNVERIFIED, and every check shows in the trace. It catches invented details, not a wrong reading of the report: you still open the cited section.

GET ONE

Pull a bug report from your own phone

1	**Enable Developer Options**	On the phone, tap Build number seven times, then turn on USB debugging.
2	**Connect and run adb**	On a computer with the Android platform tools, connect the phone.
3	**Produce the file**	adb bugreport makes a .zip (or .txt). Upload it into the chatbot.
4	**Or use the sample**	A synthetic sample_bugreport.txt is bundled, so it works before you have your own.

```
adb bugreport  mybugreport      # produces mybugreport.zip, then upload it
```

YOUR TURN

Open the chatbot and ask it something

1	**Open in Colab**	bugreport_chatbot_ui.ipynb: four steps, each with its Purpose, Input, How it works, and Output. Nothing to install on your computer.
2	**Set up, paste a free key**	Steps 1-3 install, write the app files, and ask for your free Groq key (the same as Labs 01-06). Or use local Ollama, no key.
3	**Load a report**	Step 4 opens the chat. Use the bundled sample, or upload your own adb bugreport.
4	**Ask away**	What phone is this? Which apps? Any errors in the log?
5	**Check the source**	Open the cited section and confirm it. An UNVERIFIED mark means the check failed: verify it yourself.

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Ask the report. Check the section.

Same agent idea as the device-image lab, pointed at a bug report you can pull from your own phone, and every answer still cites its source.

